

Conjecture 00000007685 is false: the leading coefficient of J_q is not $\Gamma(1/4)^2/(2\pi^{3/2})$

Statement. For $q \in (0, 1)$ let $\theta(w) = \sum_{n \in \mathbb{Z}} q^{n^2} w^n$, $T(z) = \theta(qz)/\theta(z)$ and

$$J_q = \int_0^1 T(z) d_q z = (1 - q) \sum_{n \geq 0} q^n T(q^n)$$

(Jackson measure $(1 - q) \sum z q^n f(z q^n)$ at $z = 1$). The conjecture asserts (i) J_q is transcendental for algebraic $q \in (0, 1)$ and (ii) the leading coefficient of the Puiseux expansion of J_q in $q - 1$ is $c = \Gamma(1/4)^2/(2\pi^{3/2}) \approx 1.18034$.

Reading. Clause (ii) is formalised as: there are real α and L with $J_q/(1 - q)^\alpha \rightarrow L$ as $q \rightarrow 1^-$ and $|L| = c$. Taking $|L|$ makes the branch or sign of $(q - 1)^\alpha$ irrelevant, so this is weaker than the conjecture and refuting it refutes (ii). Clause (i) is not decided. The package proves $\neg((i) \wedge (ii))$ by refuting (ii) only.

Proof. Put $a_n = \theta(q^n)$. Substituting $k \mapsto k + 1$ gives $\theta(q^2 w) = q^{-1} w^{-1} \theta(w)$ (Lean: `theta_shift`), so $a_{n+2} = q^{-1} q^{-n} a_n$. With $b_n = q^n T(q^n) = q^n a_{n+1}/a_n$ this yields $b_{n+2} = q b_n$ (`b_step`). Let $r = \theta(q)/\theta(1)$; then $b_0 = r$ and $b_1 = 1/r$, so $b_{2m} = q^m r$, $b_{2m+1} = q^m/r$ and

$$J_q = (1 - q) \sum_m q^m (r + 1/r) = r + \frac{1}{r} \quad (\text{J_eq}).$$

Termwise, $q^{k^2} \leq q^{k^2+k} + q^{(k-1)^2+(k-1)}$ and $q^{k^2+k} \leq q^{k^2} + q^{(k+1)^2}$; summing over $k \in \mathbb{Z}$ (after the shifts $k \mapsto k \pm 1$) gives $\theta(1) \leq 2\theta(q)$ and $\theta(q) \leq 2\theta(1)$, i.e. $1/2 \leq r \leq 2$. Summability of q^{k^2} is proved by comparison with $q^{|k|}$; that of q^{k^2+k} from the second inequality. Hence $2 \leq J_q \leq 4$ on $(0, 1)$ (`J_bounds`). These bounds are all the disproof uses. (Numerically J_q approaches 2 as $q \rightarrow 1^-$, but this is neither proved here nor needed.)

Now suppose $J_q/(1 - q)^\alpha \rightarrow L$. If $\alpha > 0$, then $J_q = (J_q/(1 - q)^\alpha)(1 - q)^\alpha \rightarrow L \cdot 0 = 0$, contradicting $J_q \geq 2$. If $\alpha = 0$, $L \geq 2$. If $\alpha < 0$, $0 \leq J_q(1 - q)^{-\alpha} \leq 4(1 - q)^{-\alpha} \rightarrow 0$, so $L = 0$. Finally $0 < c < 2$: convexity of Γ between $\Gamma(1) = \Gamma(2) = 1$ gives $\Gamma(5/4) \leq 1$, so $\Gamma(1/4) = 4\Gamma(5/4) \leq 4$, and $(\pi^{3/2})^2 = \pi^3 > 27$ gives $\pi^{3/2} > 4$, so $c \leq 16/(2\pi^{3/2}) < 2$ (`c_bounds`). So $|L| = c$ is impossible in all three cases.

Lean. File `lean/Conjecture7685/Basic.lean` (compiled with Lean 4.33.1, Mathlib v4.33.1). Definitions `theta`, `T`, `J`, `c`, `TranscConj`, `LeadCoeffConj`, `Conjecture`, `Claim = ¬Conjecture`. Main theorem `C7685.main` : `Claim`. Auxiliary: `J_eq`, `J_bounds`, `c_bounds`.

Conventions. θ is the Lean `tsum` over $\mathbb{Z}$ (summable for $q \in (0, 1)$, proved); powers q^{n^2} , w^n are integer powers of reals; $\mathbb{N}$ starts at 0. Not refuted: clause (i), transcendence of J_q .

Axiom audit. `#print axioms C7685.main`: see `verification/` produced by `make_pr.py`; only `propxt`, `Classical.choice`, `Quot.sound` are allowed and checked there.